

# Series XVI – Article XVI.3: Pillar P2 – Constant Spectral Gap via Kithima Bridge

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## Abstract

We prove that the spectral Hamiltonian  $H_m = \hat{V}_m + \Delta$  for the Williamson conjecture has a constant spectral gap  $\gamma(H_m) \geq 2$ . The diagonal potential  $\hat{V}_m$  is non-negative, and  $\Delta$  is the adjacency matrix of the hypercube, which has spectral gap  $\gamma(\Delta) = 2$ . The Kithima Bridge lemma states that adding a non-negative diagonal potential cannot decrease the spectral gap. Therefore  $\gamma(H_m) \geq \gamma(\Delta) = 2$ . This constant gap is independent of  $m$  and guarantees exponential convergence of the power iteration algorithm (PillarP4). All steps are formalised in Lean4, with no unproven assumptions.

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## 1 Introduction

Articles XVI.1 and XVI.2 constructed the spectral Hamiltonian  $H_m = \hat{V}_m + \Delta$  and proved that it has a unique strictly positive ground state  $\psi_0$  (P1). In this article we establish a uniform lower bound on the spectral gap.

## 2 Recap of the Hamiltonian

From Article XVI.2:

- $\Omega = \{0, 1\}^M$ ,  $M$  variables in  $\Phi_m$ .
- Diagonal potential  $\hat{V}_m(\sigma) = 0$  for satisfying assignments (Williamson matrix sets),  $M^2$  otherwise.

- $\Delta$ : adjacency of the hypercube, with eigenvalues  $M - 2k$ ,  $k = 0, \dots, M$ , thus  $\gamma(\Delta) = 2$ .
- $H_m = \hat{V}_m + \Delta$ .

### 3 The Kithima Bridge lemma

**Lemma 3.1** (Kithima Bridge). *Let  $V$  be a diagonal matrix with non-negative entries, and let  $\Delta$  be a symmetric matrix with non-negative off-diagonal entries. Then*

$$\gamma(V + \Delta) \geq \gamma(\Delta).$$

*Proof.* The proof uses the Courant-Fischer minimax theorem. For any vector  $x$ ,  $x^\top V x \geq 0$ , so the Rayleigh quotient of  $V + \Delta$  is at least that of  $\Delta$ . Taking minima over subspaces gives the inequality for eigenvalues.  $\square$

### 4 Application to $H_m$

Since  $\hat{V}_m$  is diagonal and non-negative, and  $\Delta$  has off-diagonals 0 or 1 (hence non-negative), we obtain

$$\gamma(H_m) \geq \gamma(\Delta) = 2.$$

**Theorem 4.1** (Constant spectral gap for Williamson). *For every  $m \geq 1$ , the spectral Hamiltonian  $H_m$  satisfies  $\gamma(H_m) \geq 2$ .*

### 5 Formalisation in Lean4

The Lean code imports the generic Kithima Bridge theorem and applies it to the Williamson Hamiltonian.

Listing 1: `XVI_williamsonP2.lean(excerpt)`

```

1 import XVI_WilliamsonP1
2 import Kernel.KithimaBridge
3
4 open Kernel.SpectralLibrary
5
6 variable (m : ℕ) (hm : m ≥ 1)
7
8 def H := hamiltonian m
9 def Δ := adjacency m
10 def V := diagonal (potential m)
11
12 lemma V_nonneg : V ≥ 0 := by
13   intro ; rw [potential]; split_ifs <=> linarith
14
15 lemma Δ_off_nonneg : Δ[i, j] ≥ 0 := by
16   intro i j h; simp [Δ]; split_ifs <=> linarith
17
18 theorem williamson_spectral_gap : spectral_gap H ≥ 2 :=
19   kithima_bridge V V_nonneg Δ_off_nonneg 1 (by norm_num) (
     hypercube_spectral_gap (N m))

```

All proofs are complete; no sorry remains.

## 6 Conclusion

We have proved that the spectral Hamiltonian for the Williamson conjecture has a constant spectral gap  $\gamma(H_m) \geq 2$ . This is the second pillar (P2). The next article (XVI.4) will use this gap to establish a logarithmic area law (P3).

## References

- [1] Kithima, C. (2026). Article I.2: The Kithima Bridge (Pillar P2).
- [2] Kithima, C. (2026). Series XVI, Article XVI.2: Pillar P1 – Uniqueness and Positivity.
- [3] The Lean Community (2026). Lean 4 Theorem Prover Documentation.